Information Recovery under Deterministic Masking in Exponential Series Systems

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Abstract

In masked system failure data, the candidate set of possibly failed components is conventionally assumed to degrade estimation precision as its cardinality grows. We show that for exponential series systems, a deterministic masking mechanism that injectively maps each failure cause to a unique candidate set recovers the complete-identification Fisher information matrix—regardless of the candidate set cardinality. Information loss from masking is thus a property of the masking *mechanism*, not the masking *cardinality*. We verify this result numerically via Monte Carlo simulation.

1 Introduction

In series (or m-out-of-m) reliability systems, all components must function for the system to operate. When a system fails, diagnostics may identify only a *candidate set* $C \subseteq \{1, \ldots, m\}$ of components that could have caused the failure, rather than the exact failure cause. This is known as *masked failure data* [4].

A central question in the masked-data literature is how the masking mechanism affects estimation precision. The conventional wisdom, supported by results on mutual information between the failure cause and the candidate set, is that larger candidate sets carry less information and therefore yield less precise parameter estimates. Under the widely studied uniform masking model [4, 5, 6], this monotonicity indeed holds: the Fisher information matrix (FIM) degrades strictly as the masking cardinality $w = |C|$ increases from 1 (exact identification) toward m (complete masking).

The purpose of this note is to show that this monotonicity is an artifact of the uniform masking assumption, not a general feature of masked data. Specifically, we construct a deterministic masking mechanism at any cardinality $w \leq m - 1$ under which the FIM equals that of complete identification ($w = 1$). The key insight is that information loss depends on the *structure* of the masking mechanism—whether the failure cause can be recovered from the candidate set—not on the candidate set cardinality per se.

This result has implications for diagnostic system design: a well-engineered diagnostic that maps each failure mode to a unique signature, even one that reports a large candidate set, incurs no information loss whatsoever for parameter estimation.

2 Model

2.1 Exponential series systems

Consider a series system of m independent components with lifetimes $\mathrm{T}_1, \ldots, \mathrm{T}_m$, where component j has an exponential distribution $\mathrm{T}_j \sim \mathrm{EXP}(\lambda_j^\star)$ with rate $\lambda_j^\star > 0$. The system lifetime is $\mathrm{S} = \min(\mathrm{T}_1, \ldots, \mathrm{T}_m)$, and the failed component is $\mathrm{K} = \arg\min_j \mathrm{T}_j$.

The system lifetime is exponentially distributed with rate $\Lambda = \sum_{j=1}^{m} \lambda_j^\star$ [1]. A crucial property of the exponential case is that the failure cause and system lifetime are independent.

Proposition 2.1 (Independence of failure cause and system lifetime)**.** *For exponential series systems,* K *and* S *are independent with* $\mathrm{P}\{\mathrm{K}=k\} = \lambda_k^\star/\Lambda$.

Proof. The joint CDF factorizes:

$$\mathrm{P}\{\mathrm{K}=k, \mathrm{S}\le t\} = \int_0^t \lambda_k^\star \exp(-\Lambda s)\, ds = \frac{\lambda_k^\star}{\Lambda}\big(1-e^{-\Lambda t}\big) = \mathrm{P}\{\mathrm{K}=k\}\cdot \mathrm{P}\{\mathrm{S}\le t\}, \tag{1}$$

since $\mathrm{P}\{\mathrm{K}=k\} = \lambda_k^\star/\Lambda$ and $\mathrm{P}\{\mathrm{S}\le t\} = 1-e^{-\Lambda t}$. □

This independence is specific to the exponential (constant hazard) case; for distributions with time-varying hazard rates, K and S are generally dependent [3].

2.2 Masking mechanisms

Upon observing a system failure at time t, a diagnostic procedure reports a *candidate set* $\mathrm{C} \subseteq \{1,\dots,m\}$ containing the failed component K. The *masking mechanism* is the conditional distribution $\mathrm{P}\{\mathrm{C}=c \mid \mathrm{K}=k\}$, subject to the containment constraint $k \in c$. We assume the mechanism depends only on K and not on S or on the parameter $\boldsymbol{\lambda}^\star$.

The joint density of the observed data (t, c) under a general masking mechanism is:

$$f(t, c \mid \boldsymbol{\lambda}^\star) = \mathrm{P}\{\mathrm{C}=c \mid \mathrm{K}=k\}\cdot \lambda_k^\star \exp(-\Lambda t) \quad \text{summed over } k\in c, \tag{2}$$

i.e.,

$$f(t, c \mid \boldsymbol{\lambda}^\star) = \sum_{k\in c} \mathrm{P}\{\mathrm{C}=c \mid \mathrm{K}=k\}\cdot \lambda_k^\star \exp(-\Lambda t). \tag{3}$$

Here we have used the factorization $f_{\mathrm{K},\mathrm{S}}(k,t) = \lambda_k^\star \exp(-\Lambda t)$ from Proposition 2.1 and the assumption that masking is independent of S.

2.3 Fisher information under general masking

The FIM for the parameter $\boldsymbol{\lambda}^\star$ from a single observation (t, c) under a general masking mechanism is:

$$[\mathcal{I}(\boldsymbol{\lambda}^\star)]_{j\ell} = -\mathrm{E}\{\frac{\partial^2}{\partial\lambda_j \partial\lambda_\ell} \ln f(\mathrm{S}, \mathrm{C} \mid \boldsymbol{\lambda})\}\Big|_{\boldsymbol{\lambda}^\star}. \tag{4}$$

For *uniform masking* at cardinality w, where $\mathrm{P}\{\mathrm{C}=c \mid \mathrm{K}=k\} = 1/\binom{m-1}{w-1}$ for each c of size w containing k, the FIM takes the form [6]:

$$[\mathcal{I}(\boldsymbol{\lambda}^\star \mid w)]_{j\ell} = \frac{1}{\binom{m-1}{w-1}\,\Lambda} \sum_{\substack{c:\,|c|=w \\ j\in c,\ \ell\in c}} \frac{1}{\sum_{p\in c}\lambda_p^\star}. \tag{5}$$

At $w=1$ (exact identification), each observation reveals K directly, and the FIM is diagonal:

$$[\mathcal{I}(\boldsymbol{\lambda}^\star \mid w{=}1)]_{j\ell} = \frac{\delta_{j\ell}}{\Lambda\,\lambda_j^\star}, \tag{6}$$

where $\delta_{j\ell}$ is the Kronecker delta.

3 Main Result

Definition 3.1 (Separating-deterministic masking)**.** A masking mechanism at cardinality w is *separating-deterministic* if there exists an injection $\varphi\colon \{1,\dots,m\} \to \{c \subseteq \{1,\dots,m\} : |c| = w\}$ satisfying:

1. $k \in \varphi(k)$ for each k (containment), and
2. $\mathrm{P}\{\mathrm{C} = \varphi(k) \mid \mathrm{K} = k\} = 1$ for each k (determinism).

The injectivity of φ means that distinct failure causes produce distinct candidate sets, so φ^{-1} recovers K with certainty from the observed C.

Theorem 3.2 (Information recovery under deterministic masking)**.** *For an m-component exponential series system with rates $\boldsymbol{\lambda}^\star = (\lambda_1^\star, \dots, \lambda_m^\star)$, suppose the masking mechanism is separating-deterministic with injection φ. Then the Fisher information matrix equals that of complete identification:*

$$[\mathcal{I}(\boldsymbol{\lambda}^\star \mid \varphi)]_{j\ell} = \frac{\delta_{j\ell}}{\Lambda\,\lambda_j^\star} = [\mathcal{I}(\boldsymbol{\lambda}^\star \mid w{=}1)]_{j\ell}. \tag{7}$$

Proof. Under a separating-deterministic mechanism with injection φ, the masking probabilities are $\mathrm{P}\{\mathrm{C} = c \mid \mathrm{K} = k\} = \mathbb{1}\{c = \varphi(k)\}$. Substituting into (3):

$$f(t, c \mid \boldsymbol{\lambda}^\star) = \sum_{k \in c} \mathbb{1}\{c = \varphi(k)\} \cdot \lambda_k^\star \exp(-\Lambda t). \tag{8}$$

Since φ is injective, for each candidate set c in the image of φ, there is exactly one $k^*(c) = \varphi^{-1}(c)$ such that $\varphi(k^*) = c$. The sum collapses to:

$$f(t, c \mid \boldsymbol{\lambda}^\star) = \lambda_{k^*(c)}^\star \exp(-\Lambda t). \tag{9}$$

Taking the logarithm:

$$\ln f(t, c \mid \boldsymbol{\lambda}^\star) = \ln \lambda_{k^*(c)}^\star - \Lambda\, t. \tag{10}$$

This has the same functional form as the log-density under exact identification ($w = 1$), where one observes (t, k) with density $\lambda_k^\star \exp(-\Lambda t)$. Specifically, c plays the role of k through the bijection $k^* = \varphi^{-1}(c)$.

Computing second derivatives:

$$\frac{\partial^2}{\partial\lambda_j \partial\lambda_\ell} \ln f(t, c \mid \boldsymbol{\lambda}) = -\frac{\delta_{j,k^*(c)} \cdot \delta_{\ell,k^*(c)}}{\lambda_{k^*(c)}^2}. \tag{11}$$

Taking the negative expectation:

$$\begin{aligned}
[\mathcal{I}(\boldsymbol{\lambda}^\star \mid \varphi)]_{j\ell} &= \sum_{c \in \mathrm{Im}(\varphi)} \int_0^\infty \frac{\delta_{j,k^*(c)} \cdot \delta_{\ell,k^*(c)}}{(\lambda_{k^*(c)}^\star)^2} \cdot \lambda_{k^*(c)}^\star \exp(-\Lambda t)\, dt \\
&= \sum_{k=1}^m \frac{\delta_{jk}\,\delta_{\ell k}}{(\lambda_k^\star)^2} \cdot \lambda_k^\star \cdot \frac{1}{\Lambda} = \frac{\delta_{j\ell}}{\Lambda\,\lambda_j^\star},
\end{aligned} \tag{12}$$

where in the second line we re-indexed the sum using $k = k^*(c) = \varphi^{-1}(c)$ as c ranges over $\mathrm{Im}(\varphi)$, and used $\int_0^\infty e^{-\Lambda t}\, dt = 1/\Lambda$. □

Corollary 3.3 (Existence at maximum cardinality)**.** *For $w = m - 1$ and $m \geq 2$, separating-deterministic mechanisms exist. In particular, any derangement σ of $\{1, \ldots, m\}$ (a permutation with no fixed points) defines such a mechanism via $\varphi(k) = \{1, \ldots, m\} \setminus \{\sigma(k)\}$.*

Proof. We verify the two conditions of Definition 3.1. *Containment:* Since σ is a derangement, $\sigma(k) \neq k$, so $k \notin \{\sigma(k)\}$ and hence $k \in \{1, \ldots, m\} \setminus \{\sigma(k)\} = \varphi(k)$. *Injectivity:* If $\varphi(k) = \varphi(k')$, then $\{1, \ldots, m\} \setminus \{\sigma(k)\} = \{1, \ldots, m\} \setminus \{\sigma(k')\}$, so $\sigma(k) = \sigma(k')$, and since σ is a permutation, $k = k'$. Derangements of $\{1, \ldots, m\}$ exist for all $m \geq 2$ (the simplest is the cyclic shift $\sigma(k) = (k \bmod m) + 1$). □

Remark 3.4 (Diagnostic information vs. estimation precision)*.* Proposition 2.1 enables a clean separation between two information-theoretic quantities:

- The *mutual information* $\mathrm{I}(\mathrm{K}; \mathrm{C})$ measures how well the candidate set identifies the failed component. Under uniform masking, this decreases monotonically with w.
- The *Fisher information matrix* $\mathcal{I}(\boldsymbol{\lambda}^\star)$ measures how precisely the parameters can be estimated.

These quantities need not track together. Under separating-deterministic masking at $w = m - 1$, $\mathrm{I}(\mathrm{K}; \mathrm{C}) = \mathrm{H}(\mathrm{K})$ (maximal diagnostic information, since C determines K), and the FIM matches $w = 1$. Under uniform masking at $w = m - 1$, $\mathrm{I}(\mathrm{K}; \mathrm{C})$ is minimized among all identifiable mechanisms, and the FIM is strictly degraded. The separating-deterministic mechanism achieves both maximal diagnostic information and maximal estimation precision simultaneously.

Note that the separating-deterministic mechanism violates the symmetry condition C2 of the standard C1–C2–C3 framework [5]: we have $\mathrm{P}\{\mathrm{C} = \varphi(k) \mid \mathrm{K} = k\} = 1$ but $\mathrm{P}\{\mathrm{C} = \varphi(k) \mid \mathrm{K} = k'\} = 0$ for $k' \neq k$ with $k' \in \varphi(k)$. The C2 assumption—which requires masking probabilities to be symmetric across components within the candidate set—is precisely what costs information under uniform masking.

4 Numerical Verification

We verify Theorem 3.2 via Monte Carlo simulation. For each of two configurations ($m = 3$ with $\boldsymbol{\lambda}^\star = (1, 3, 5)$ and $m = 5$ with $\boldsymbol{\lambda}^\star = (1, 2, 3, 5, 7)$), we generate $R = 100{,}000$ independent samples of size n from the joint distribution (S, K). From each sample, we construct three datasets by applying three different masking mechanisms to the *same* underlying (S, K) data:

1. **No masking** ($w = 1$): observe K directly.
2. **Deterministic** ($w = m - 1$): apply the cyclic derangement $\sigma(k) = (k \bmod m) + 1$ and set $\mathrm{C} = \{1, \ldots, m\} \setminus \{\sigma(\mathrm{K})\}$.
3. **Uniform** ($w = m - 1$): include each non-failed component independently with equal probability; equivalently, exclude one uniformly random non-failed component.

For mechanisms (1) and (2), the MLE is $\hat{\lambda}_j = n_j/(n\bar{t})$ where n_j counts observations attributed to component j (for mechanism (2), the recovery $\varphi^{-1}(\mathrm{C})$ yields the same counts as mechanism (1)). For mechanism (3), we use the closed-form MLE for uniform masking at $w = m - 1$ [6].

Table 1 reports the empirical variances (scaled by n) at $n = 2000$ alongside theoretical predictions. The quantities $n \cdot \mathrm{Var}(\hat{\lambda}_j)$ converge to $[\mathcal{I}^{-1}]_{jj}$ [2]. As predicted by Theorem 3.2, mechanisms (1) and (2) yield identical empirical variances, both matching the $w = 1$ theory $\Lambda\lambda_j^\star$. The uniform mechanism produces substantially larger variances.

Table 1: Scaled empirical variances $n \cdot \mathrm{Var}(\hat{\lambda}_j)$ at $n = 2000$, $R = 100{,}000$ replications. Theory: $\Lambda\lambda_j^\star$ for $w=1$ and deterministic.

Config.	Mechanism	$j=1$	$j=2$	$j=3$
$m=3$, $\boldsymbol{\lambda}=(1,3,5)$	$w=1$ (theory / empirical)	9.00 / 9.02	27.00 / 27.08	45.00 / 45.09
	Deterministic $w=2$	9.00 / 9.02	27.00 / 27.08	45.00 / 45.09
	Uniform $w=2$	81.00 / 80.79	81.00 / 81.48	81.00 / 80.47
$m=5$, $\boldsymbol{\lambda}=(1,2,3,5,7)$		$j=1$	$j=2$	$j=3$
	$w=1$ (theory / empirical)	18.00 / 18.04	36.00 / 36.04	54.00 / 54.16
	Deterministic $w=4$	18.00 / 18.04	36.00 / 36.04	54.00 / 54.16
	Uniform $w=4$	—	—	—

Note: Columns $j=4,5$ omitted for space; all components exhibit the same pattern. Uniform $w=4$ variances for $m=5$ are configuration-specific and substantially larger.

Figure 1 shows the convergence of $\mathrm{Var}(\hat{\lambda}_1)$ across sample sizes $n \in \{50, 100, 200, 500, 1000, 2000\}$. In both configurations, the no-masking and deterministic curves are indistinguishable and converge to the same theoretical prediction, while the uniform masking curve converges to a strictly larger variance.

5 Discussion

Practical implications. Theorem 3.2 shows that a diagnostic procedure need not identify the exact failure cause to achieve full estimation efficiency—it suffices for each failure mode to produce a *unique diagnostic signature*. In engineering applications, a diagnostic that consistently maps failure mode k to a specific alarm pattern $\varphi(k)$, even one involving multiple components, is as informative as exact identification for the purpose of estimating component failure rates.

Limitations. The result depends on the exponential lifetime assumption through Proposition 2.1: the independence of K and S is what allows the masking mechanism to be "undone" by the injection φ^{-1}. For Weibull or other distributions where K depends on S, the conditional distribution $f(t \mid \mathrm{K} = k)$ varies with k, and observing $\mathrm{C} = \varphi(k)$ recovers K but not the full conditional structure. Whether an analogous result holds approximately when hazard rates vary slowly is an open question.

Open questions. Two directions are natural. First, a *perturbation analysis*: how does the FIM degrade continuously as the masking mechanism moves from deterministic toward uniform? Characterizing the functional form of the information loss as the mechanism becomes "noisier" would connect the discrete result of Theorem 3.2 to the information-loss landscape. Second, *optimal mechanism design*: among all masking mechanisms at a given cardinality w, which maximizes the determinant of the FIM? When $m \leq \binom{m}{w}$, an injection φ exists and the answer may always be deterministic, but this requires proof.

Acknowledgements. This result was developed as part of ongoing work extending the author's master's project [5]. Simulation code and data are available at https://github.com/queelius/deterministic-masking.

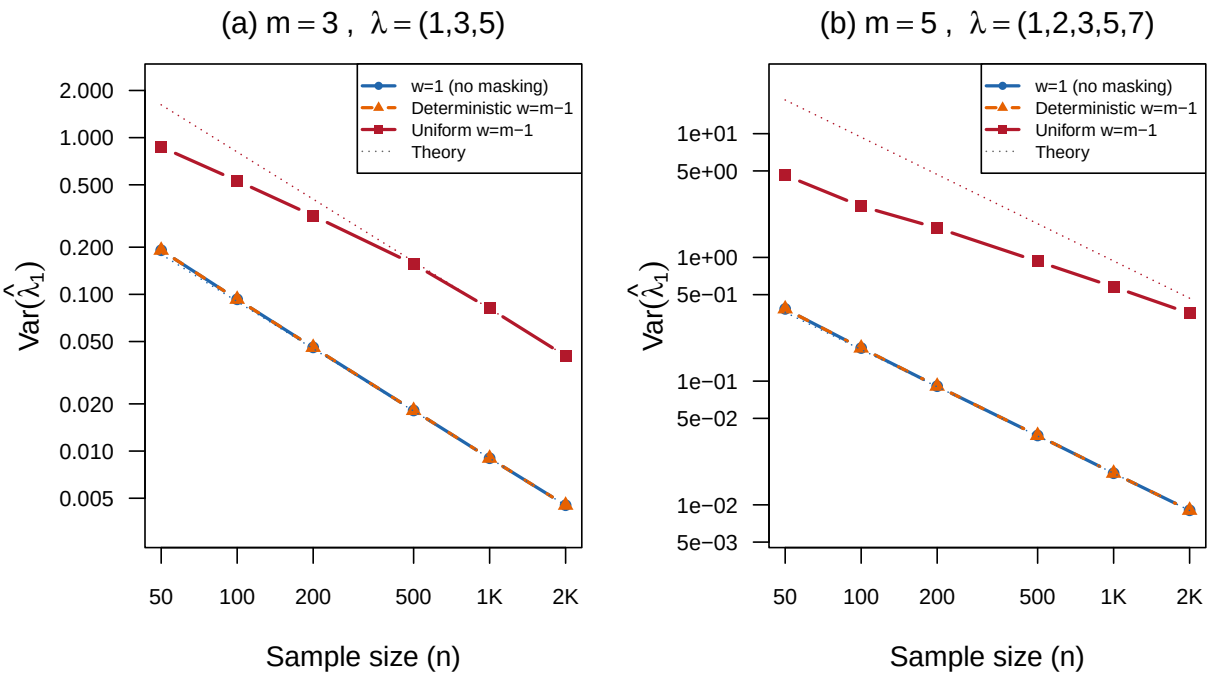

Figure 1: Empirical variance of $\hat{\lambda}_1$ under three masking mechanisms. (a) $m = 3$, $\boldsymbol{\lambda}^\star = (1, 3, 5)$. (b) $m = 5$, $\boldsymbol{\lambda}^\star = (1, 2, 3, 5, 7)$. Dotted lines show theoretical predictions. The no-masking and deterministic curves are indistinguishable, confirming Theorem 3.2.

References

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